

Math Foundations for Decision and Data Sciences

ISE 201

Fall 2025 Section 33 Fully Online 3 Unit(s) 10/07/2025 to 12/16/2025 Modified 10/03/2025

Contact Information

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Course Information

Linear algebra and matrix operations in statistics and optimization. Statistical concepts and techniques that are important for decision and data sciences. Also, basic optimization techniques useful for statistics.

Course Description and Requisites

Linear algebra and matrix operations in statistics, optimization and artificial intelligence. Statistical concepts and techniques important for decision and data sciences, with motivation by real-world examples. Basic optimization techniques useful for statistics, classification and neural networks.

Prerequisite: Graduate Standing in Engineering

Letter Graded

Classroom Protocols

Our four hours Tuesday weekly sessions are online from 5:30pm through 9:30pm (Pacific Time) on Tuesdays, starting October 7th, and ending December 16th. Since November 11th is Veterans Day a Federal Holiday in the US. where SJSU campus is closed, that week we will meet on Wednesday November 12.

For each session, our learning will be organized into Four sections. Each section will have a Lecture portion (~20 minutes), team exercise for you to work on in a breakout room (~15 minutes), teams reporting their outcomes (~10 minutes), and a short break (5-10 minutes). These are rough time estimates.

Please make sure that your video and audio works well. It is important that we all have our video on at all times (except during breaks).

In closing, please put best effort to be on time for class. There are times when you can't help it, and I will understand. But as a statistician, I can verify that if a person is late on a regular basis, they probably don't respect other people's time. Please respect the time of your classmates and be on time. THANK YOU

Course Goals

Develop student knowledge of linear algebra and matrix operations in statistics, and optimization.

Develop student knowledge of statistical concepts and techniques important for decision and data sciences.

Develop student knowledge of basic optimization techniques useful for statistics, classification and neural networks.

Course Learning Outcomes (CLOs)

The following are the Course Learning Outcomes (CLO). The class work/assignments that most support that CLO are highlighted in *italic bold*.

CLO 1. To use multi-dimensional arrays to represent data of independent variables and responses, including to learn about descriptive statistics and graphical methods to summarize data using statistical software (Python). (***Week One, Homework 01 and Project***)

CLO 2. Review linear algebra topics and to manipulate arrays and matrices to discover variables-responses relationships. (***Weeks Two & Three, Homework 02, Quiz 01 and Project***)

CLO 3. To learn, master and explain basic probability concepts and its use in statistical inference. (***Weeks Four & Five, Homework 03, Quiz 02 and Project***)

CLO 4. To learn, master and explain basic concepts of statistical inference. (***Weeks Six & Seven, Homework 04, Quiz 03 and Project***)

CLO 5. To formulate General Linear Models, including regression for quantitative variables and analysis of variance for discrete variables and know and explain usefulness and limitations. (***Weeks Eight Nine, Homework 04, Quiz 03 & Ten and Project***)

CLO 6. To use statistical software to analyze data, reach statistical valid conclusions and explain them to peers and management. (*throughout class and Project*)

CLO 7. To understand and explain the usefulness and limitations of basic optimization techniques for linear and nonlinear regression. (*Weeks Eight Nine & Ten, Homework 05, Quiz 04 and Project*)

Course Materials

Access to Python IDE on the computers they will be using for this class

Applied Statistics and Probability for Engineers. D. C. Montgomery and G. Runger, 7th Edition, Wiley, 2018.

Math for Deep Learning. R. T. Kneusel, O'Reilly, 2021.

The following book is **optional** (based on your level of knowledge with Python):

Python for Data Analysis. W. McKinney, 3rd Edition, 2022

We will also use 4 chapters from the following book (**don't obtain until we discuss in class**):

Introduction to Operations Research. F. S. Hillier and G. J. Lieberman, 10th Edition, McGraw-Hill, 2015.

Course Requirements and Assignments

- 1) There are homework assignments that we will start in class that will need to be completed and turned in.
- 2) There are four quizzes that occur after we complete each of the following discussions: Linear Algebra, Probability, Statistics, and Optimization. The first 3 quizzes will be take-home, and the last one will be in-class.
- 3) This course has a **final project**. where you will write a technical blog post related to a data analysis project you will work on piecemeal over the course of the semester. You will have 3 deliverables: Proposal (week 03), EDA (week 07), Podcast (week 11)

Publishing this blog is optional. Please share the link if you do!

There are plenty of open datasets available for you to use for this project:

<https://datasetsearch.research.google.com>

<https://www.data.gov/>

<https://www.dataquest.io/blog/free-datasets-for-projects/>

Grading Information

5 Homework Assignments

20%

4 Quizzes (taken in class)	40%
Project Proposal	10%
Project EDA	10%
Project Podcast	20%

Final Grading Cut-offs:

A+ = 96.5-100%	B = 82.5-86.4%	C- = 67.5-70.5%	F Less than 55.5%
A = 92.5-96.4%	B- = 78.5-82.4%	D+ = 65.5-67.4%	
A- = 88.5-92.4%	C+ = 76.5-78.4%	D = 60.5-65.4%	
B+ = 86.5-88.4%	C = 70.5-76.4%	D- = 55.5-60.4%	

University Policies

Per [University Policy S16-9 \(PDF\)](http://www.sjsu.edu/senate/docs/S16-9.pdf) (<http://www.sjsu.edu/senate/docs/S16-9.pdf>), relevant university policy concerning all courses, such as student responsibilities, academic integrity, accommodations, dropping and adding, consent for recording of class, etc. and available student services (e.g. learning assistance, counseling, and other resources) are listed on the [Syllabus Information](https://www.sjsu.edu/curriculum/courses/syllabus-info.php) (<https://www.sjsu.edu/curriculum/courses/syllabus-info.php>) web page. Make sure to visit this page to review and be aware of these university policies and resources.

Course Schedule

Date	Week	Book	Chapter(s)	Topic	Assignment(s)
October 7	1	Montgomery	6	Descriptive Statistics, EDA	
October 14	2	Kneusel	5	Linear Algebra	
October 21	3	Kneusel	6	Linear Algebra	Homework 01 Proposal
October 28	4	Montgomery	2,3,4,5	Probability	Homework 02 Quiz 01
November 4	5	Montgomery	7,8	Probability	

November 12	6	Montgomery	9,10	Statistics	Homework 03 Quiz 02
November 18	7	Montgomery	11,12,13	Statistics	EDA Due
November 25	8	Montgomery		Statistics	
December 2	9	Hillier	2,3	Optimization	Homework 04 Quiz 03
December 9	10	Hillier	4,5	Optimization	
December 16	11			Review	Homework 05 Quiz 4 Podcast